7 An electron is trapped in a one-dimensional infinite potential well of width L along the x-direction. It can be modelled as a standing wave described by the following wavefunction:

$$\psi(x) = \begin{cases} A\sin\left(\dfrac{n\pi x}{L}\right) & 0 \le x \le L \\ 0 & \text{everywhere else} \end{cases}$$

where n is associated with different possible energy states (n = 1, 2, 3, …)

(a) For the n = 2 state,

(i) Sketch in Figure 7.1 the wave function $\psi(x)$ of the electron.

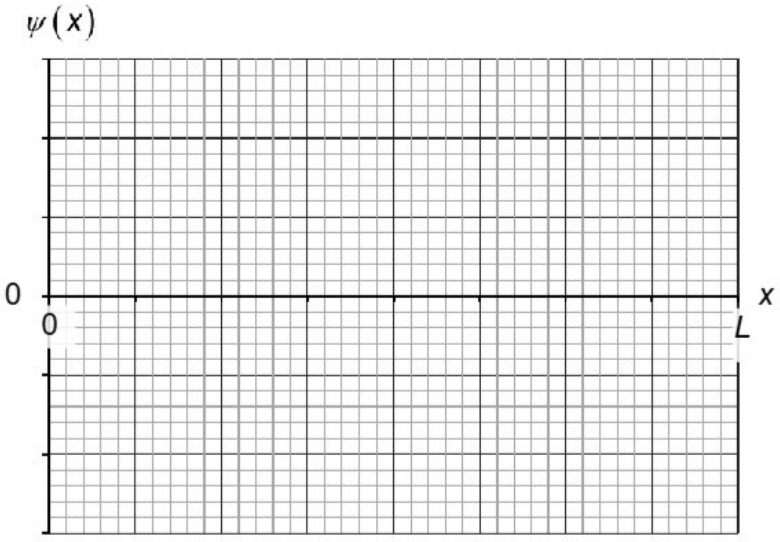


Figure 7.1

[1]

(ii) Sketch in Figure 7.2 the probability density function $|\psi(x)|^2$.

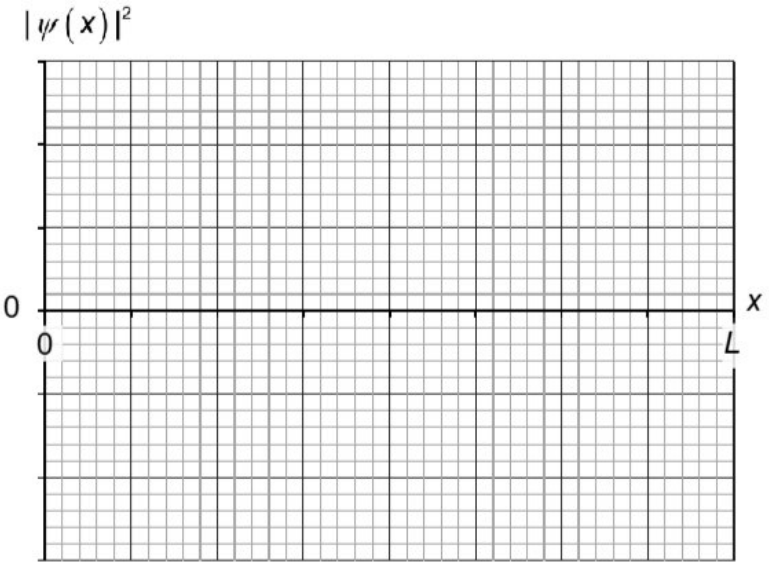


Figure 7.2

[1]

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(b) **(i)** Explain what is meant by a *normalised wave function*.

.....

..... [1]

(ii) Express the constant A in terms of L so that the wave function is normalised.

A = ... [1]

(c) A semiconductor quantum dot can be modelled as an infinite potential well.

A particular semiconductor quantum dot has an effective length L of 2.2 nm. An electron trapped in it has an effective mass of 0.15 times the mass of a free electron.

Determine the wavelength of the photon emitted when an electron in the first excited state drops to the ground state.

wavelength = ... m [3]

[Total: 7]